

Towards Robust Detection of Climate-Induced Rainfall Anomalies Using Mahalanobis Distance and Feature-Enriched Time Series

Nikhil Sanjay¹, Vedhesh Dhinakaran¹, Kopperla Bharath Reddy¹, N. Neelima², T. V. Smitha^{3*}, Vivek Venugopal²

¹Department of Computer Science and Engineering, Amrita School of Computing, Bengaluru,
Amrita Vishwa Vidyapeetham, Bengaluru, India

²Department of Electronics and Communication Engineering, Amrita School of Engineering, Bengaluru,
Amrita Vishwa Vidyapeetham, India

³Computational Science Lab, Department of Mathematics, Amrita School of Engineering, Bengaluru,
Amrita Vishwa Vidyapeetham, India

*Corresponding author: tv_smitha@blr.amrita.edu

Abstract—The frequency of hydro-meteorological extremes that is on the rise in South Asia requires the implementation of effective anomaly detection systems for successful disaster preparedness. This paper introduces a multivariate anomaly detection methodology utilizing Mahalanobis distance for identifying compound anomalies of rainfall, soil moisture, and temperature in Baksa district, Assam, from April 2023 to October 2024. A new 14-feature dataset combining VIC model rain estimation, in-situ soil moisture, and AWS temperature measurements was created. Temporal and hydrological memory features were designed, and data imputed and standardized. Mahalanobis distance combined with the Mahalanobis-Taguchi System allowed for multivariate anomaly detection with consideration for inter-feature correlations and scale differences. Validation with ASDMA disaster logs, climatological deviation thresholds, and ENSO modulation identified a detection accuracy above 82%. The method surpassed conventional univariate approaches and provides a scalable solution to real-time monitoring of regions affected by monsoons.

Index Terms—Rainfall Anomaly Detection, Mahalanobis Distance, Multivariate Time Series, Hydro-Meteorological Extremes, Mahalanobis-Taguchi System, Soil Moisture Monitoring, Climate Variability, Early Warning Systems

I. INTRODUCTION

The growing frequency and intensity of compound hydro-meteorological extremes in monsoon-dominated regions of South Asia have augmented the imperative for strong early-warning systems with the ability to identify anomalous conditions before these escalate to significant disasters. Baksa District, situated on the northern Brahmaputra floodplain at 26.7°N, 91.3°E, is a classic example of the susceptibility of such places, where more than 2,971.6 mm rainfall occurs each year but with very high variability that causes both catastrophic floods and acute droughts within the same phase of seasons. The location of the district within the monsoon-preponderant climate zone and its exposure to transboundary water flows from Bhutan’s reservoir systems forms a hydro-climatically

complicated environment where conventional univariate monitoring practices frequently struggle to detect the interlinked character of rainfall, soil moisture, and temperature anomalies. Existing operational early warning systems used by the Assam State Disaster Management Authority are based primarily on single-variable thresholds, including generally rainfall exceedance values of 100 mm per day or river stage measurements that initiate flood warnings. Such approaches in a systematic manner overlook the decisive role played by antecedent soil moisture conditions and thermal stress factors that have a direct effect on the shift of moderate rainfall to flood events or droughts. Literature shows that these univariate methods lead to elevated false alarm ratios during times of spiky rainfalls over dry conditions and, on the other hand, will miss flood events when moderate rainfall occurs over already saturated ground conditions.

II. LITERATURE SURVEY

Multivariate anomaly detection in hydro-meteorology has developed very quickly over the last few years, spurred on by demands to diagnose compound events that cannot be detected using univariate methods. Standard methods involved single variables—often rainfall—through threshold exceedances or elementary statistical aggregates. Although useful for heavy rainfall, these were unable to represent interactive processes like antecedent soil saturation and thermal stress, so missed too many events in monsoon-influenced basins like the Brahmaputra floodplain and had high false-alarm rates [1]. Standard methods involved single variables—often rainfall—through threshold exceedances or elementary statistical aggregates [2]. Early multivariate approaches used principal component analysis (PCA) to compress dimensions prior to using Euclidean-distance metrics in transformed space. PCA–Euclidean approaches enhanced detection of spatially consistent anomalies but were still scale-sensitive and relied on spherical cluster shapes, constraining their capability to describe coherent, non-spherical anomaly shapes [3]. Kernel-density estimation and

local outlier factor methods brought nonparametric flexibility to the table, allowing for detection of excursions on nonlinear manifolds, but at the expense of substantial computational costs on huge, daily time-series data sets and tending to perform poorly when faced with sparsely sampled networks of soil-moisture [4], [5].

The Mahalanobis distance (MD) became the most favored method for multivariate hydro-meteorological anomaly detection because it utilized the entire covariance structure and provided scale invariance and sensitivity to correlated deviations in features [6], [7]. Foundational studies demonstrated MD's superiority over univariate z-scores in detecting compound drought and flood precursors, reporting recall improvements from around 0.60 to above 0.80 when temperature and soil moisture were included alongside precipitation [8]. The covariance estimation technique like Minimum covariance determinant (MCD) and the M-estimator of Tyler has been incorporated in MD framework to counter the effect of outliers and heavy tailed distributions associated with hydro-climatic variables [9].

According to MD, the Mahalanobis-Taguchi System (MTS) extended empirical thresholding that is percentile-based, and the non-Gaussian data distribution is supported without parametric assumptions made to support it [10]. In the environmental setup, MTS applications achieved 92% accuracy in rainfall classification in Malaysia and F1 values of more than 0.85 in Canada in the flood-telemetry outliers detection operation of MTS applications [11]. In the environmental setting, MTS applications achieved 92-percent correctness in rainfall categorization in Malaysia and F1 values of over 0.85 in flood-telemetry outlier detection of MTS applications in Canada outliers detection operation check was also used to enhance resiliency in its operations as it approximated the uncertainty in the threshold and notified the frequency of recalibration in response to varying climate signals [12].

In the literature, complementary validation techniques emphasize the usefulness of independent event histories, such as disaster management logs, to the estimation of the practical detection ability. Monsoon climate experiments have cross-validated MD-based anomaly notifications with the official floods and droughts with more than 80 percent alignment rates of the experiment results [13]. Further validation metrics (e.g., $\pm 1.5 \sigma$ deviation of rainfall) and additional testing of ENSO teleconnection tests, demonstrate that MD-based systems find known patterns of variability of the climate (modulated) [14]. Despite these advancements, district-scale implementations that combine satellite rainfall grids, soil-moisture sensors on the ground, and the temperature records of the AWSs have loopholes in them yet thereof have been made [15]. The past studies have been largely restricted to watershed or country scale and very little emphasis has been paid on the high-resolution and local-scale early-warning studies. The paper has addressed the gap by integrating VIC rainfall, 15 cm soil-moisture probe, and AWS temperature into a 14-feature Mahalanobis-Taguchi framework to gain a validated framework to detect multivariate anomalies in data-sparse, monsoon-

prone districts.

III. METHODOLOGY

This section provides the end-to-end pipeline that was created to detect hydro-meteorological anomalies of the Baksa District, Assam, in the month of April 2023- October 2024. The paper premises on a multivariate dataset which is a combination of the NRSC VIC rainfall model output, in-situ soil moisture, and temperature data, feature engineering and Mahalanobis distance as an anomaly detector.

MTS is a generalization of the Mahalanobis distance framework to have empirical threshold selection via percentile distributions. This method is desirable when compared to parametric thresholds because it can be used in non-Gaussian hydro-climatic data and it has a better chance of being robust due to bootstrapped percentile-based calibration.

The MTS framework defines the anomaly detection threshold in a multivariate feature space through estimating the mean vector and covariance matrix of the standardized features. The Mahalanobis space was constructed based on the baseline period (2018-2022). Mahalanobis distance of the individual observation measures the departure of a given observation to normal situations of hydro-meteorology.

Anomaly threshold, denoted as τ was empirically set as the 95 th percentile of Mahalanobis distance distribution of the baseline statistics. There was the rolling aggregation statistics (e.g., rainfall and soil moisture) to minimize non-persistent noise. Also, the anomaly frequency was examined relative to the phases of the ENSO to make sure that the monsoon variability induced by climate is sensitive.

A. Study Area and Data Fusion

The Baksa District is located in the monsoon dominated Brahmaputra floodplain (26.7°N, 91.3°E), where there is a lot of rainfall and is susceptible to droughts and extreme rainfalls.

VIC-SM-T is the primary dataset and it was built through integration of:

- NRSC VIC model rainfall estimates: 5.5 km grid, water-balance consistent, satellite-based.
- Soil moisture content: Capacities measured at a depth of 15 cm with capacitance sensors installed by the Assam Agriculture Department.
- Temperature data: Daily max and min temperature Automatic Weather Stations (AWS).

The data is in 2018-2024, and 14 engineered features, where each observation is a district-day. The sample data is presented in Figure 1. The entire pipeline of data fusion as well as validation is shown by Figure 2.

B. Data Processing and Feature Engineering

1) *Data Integration and Preprocessing:* Multiple time-series datasets were integrated into a unified daily-resolution dataset at the district scale. Temporal alignment and spatial harmonization were performed to ensure consistency. Observations were aggregated at the district level to maintain uniform spatiotemporal coverage.

Missing values (less than 6%) were imputed using linear interpolation within each time series. Noise reduction was achieved through feature standardization and rolling aggregations.

2) *Feature Engineering*: A total of 14 features were engineered to capture multi-scale hydro-meteorological variability:

- **Temporal Seasonality (4 features):**
 - Cyclical transformations: month_sin, month_cos (period = 12)
 - Temporal indicators: day_of_year, week_of_year
- **Hydrological Memory (7 features):**
 - Lagged rainfall: Rainfall_lag_1d, Rainfall_lag_3d, Rainfall_lag_7d
 - Aggregated indicators: Rainfall_7d_sum, Rainfall_7d_avg, SoilMoisture_7d_avg
- **Thermal and Baseline Context (3 features):**
 - Annual_Temp: Long-term climatic baseline
 - Temp_Seasonal: Intra-annual temperature variation

3) *Standardization and Leakage Prevention*: All features were normalized using z-score standardization (mean = 0, standard deviation = 1) based only on the 2018–2022 baseline to prevent data leakage. Figure 3 shows the feature correlation matrix. Strong positive correlation is observed between SoilMoisture_7d_avg and Rainfall_7d_avg (≈ 0.95), while Temp_Seasonal and month_cos exhibit strong negative correlation (≈ -0.83).

C. Mahalanobis Distance–Based Multivariate Anomaly Detection

Mahalanobis distance (MD) is used to identify anomalous patterns across temperature, precipitation, and soil moisture.

Let X_{train} be the standardized training dataset with $p = 14$ features and n observations.

The mean vector μ and covariance matrix Σ are defined as:

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

$$\Sigma = \frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)(x_i - \mu)^T \quad (2)$$

The Mahalanobis distance for an observation x is:

$$D^2(x) = (x - \mu)^T \Sigma^{-1} (x - \mu) \quad (3)$$

The metric considers the inter-feature correlations and scale differences as well, which allow identifying the anomalies of compounds simultaneous high temperatures, low soil moisture, and rainfall shortage.

The 95th percentile of Mahalanobis distance distribution of the training data is taken as the anomaly threshold, denoted by τ . The stability of the parameter of the τ was estimated through bootstrapping.

D. Model Validation

Three validation strategies were used to evaluate the model:

- **Historical Verification**: 82% of the anomaly days (2023–2024) matched those of ASDMA on record.
- **Climatological Deviation**: The majority of the anomalies were related to precipitation deviations beyond the standard deviation of precipitation which were above and below ± 1.5 .
- **ENSO/Monsoon Modulation**: It was found that there was higher frequency of anomaly during the El Niño and La Niña phases.

The model had accuracy of 0.77, recall of 0.81 and Heidke Skill Score (HSS) of 0.62, which is better than the univariate methods of anomaly detection.

Mahalanobis distance is the most appropriate in this case since that preserves inter-feature correlations and scale differences. It is effective at synchronizing moderate, yet statistically significant, deviation in most features - to detect engaged climate anomalies such as high temperature, low ground moisture, and precipitation shortage occurring simultaneously at the same time, among others, with the original report being true [20], [21]. In the Mahalanobis-Taguchi System, the 95th percentile of Mahalanobis distances computed on the training set was used as the empirical definition of the τ anomaly threshold [22]. It is a non-parametric cutoff that is resistant to non-Gaussian hydro-climatic data distributions of types of data [23]. The stability of testing the stability of τ was done through bootstrapping. Empirical evidence: Eco-climatology literature, structural health monitoring and hydrology indicate that Mahalanobis-based detection would be more effective than univariate or Euclidean detection, in case the features have statistical dependence between them [24], [25].

E. Validation of Model

Three validity tests were utilized:

- **Historical Verification**: Anomaly days from 2023–2024 were cross-checked with actual flood and drought days for Baksa from ASDMA. Over 82% of the anomaly days coincided with known days of disaster [26].
- **Climatological Deviation**: The majority of irregular days had precipitation anomalies beyond ± 1.5 standard deviations from the 2018–2022 district average.
- **ENSO/Monsoon Modulation**: Anomaly frequency was higher in El Niño and La Niña events, as would be expected from their local impact on monsoon rainfall.

The Mahalanobis model was 0.77 accurate, 0.81 recall, and Heidke Skill Score of 0.62, better than univariate-based detectors.

IV. RESULTS AND DISCUSSIONS

The results of the Mahalanobis distance–based anomaly detection algorithm applied to the integrated VIC-SM-T dataset for Baksa District, Assam (April 2023–October 2024) are described in this section. The results are assessed in the context of anomaly properties, temporal trends, and agreement with observed hydro-meteorological events.

State	District	Date	Year_rain	Month_rain	Avg_rainfall	Agency_name_rain	Avg_smbl_at15cm	Year	Annul_Temp	Temp_Seasonal	Rainfall_lag_1d	Rainfall_lag_3d	Rainfall_lag_7d	Rainfall_lag_15d	Rainfall_lag_30d	SoilMoisture_7d_avg	day_of_year	week_of_year	month_sin	month_cos
Assam	Baksa	31-07-2018	2018	7	13.87248454	NRSC VIC MODEL	28.5070212	2018	25.9	28.28	13.38430056	46.85087704	12.47735555	141.7266516	20.2466452	28.14189637	212	31	-0.5	-0.866025404
Assam	Baksa	01-08-2018	2018	8	10.95613136	NRSC VIC MODEL	28.421279	2018	25.9	28.28	13.87248454	3.253874344	17.623809396	135.054089	19.293527	28.23892827	213	31	-0.866025404	-0.5
Assam	Baksa	02-08-2018	2018	8	9.641263977	NRSC VIC MODEL	28.427208	2018	25.9	28.28	10.95613136	13.38430056	7.582375972	137.115777	19.58765386	28.42368291	214	31	-0.866025404	-0.5
Assam	Baksa	03-08-2018	2018	8	0.092122979	NRSC VIC MODEL	26.8654198	2018	25.9	28.28	9.641263977	46.85087704	39.1546452	98.6510548	14.00729554	28.22312394	215	31	-0.866025404	-0.5
Assam	Baksa	04-08-2018	2018	8	9.458151908	NRSC VIC MODEL	26.8840971	2018	25.9	28.28	0.092122979	10.95613136	46.85087704	60.65832967	8.665475667	27.9294987	216	31	-0.866025404	-0.5
Assam	Baksa	05-08-2018	2018	8	3.195123295	NRSC VIC MODEL	26.1929195	2018	25.9	28.28	9.458151908	9.641263977	3.253874344	60.59957862	8.65708266	27.61983689	217	31	-0.866025404	-0.5
Assam	Baksa	06-08-2018	2018	8	3.28657928	NRSC VIC MODEL	26.0153138	2018	25.9	28.28	3.195123295	0.092122979	13.38430056	50.50187334	7.21451048	27.33046549	218	32	-0.866025404	-0.5
Assam	Baksa	07-08-2018	2018	8	17.49198273	NRSC VIC MODEL	27.2081921	2018	25.9	28.28	3.28657928	9.458151908	13.87248454	54.12135553	7.73162218	27.14401847	219	32	-0.866025404	-0.5
Assam	Baksa	08-08-2018	2018	8	6.343723763	NRSC VIC MODEL	27.1793771	2018	25.9	28.28	17.49198273	3.195123295	10.95613136	49.50894793	7.072706847	26.96750391	220	32	-0.866025404	-0.5
Assam	Baksa	09-08-2018	2018	8	3.027707781	NRSC VIC MODEL	26.2686502	2018	25.9	28.28	6.343723763	3.28657928	9.641263977	42.89539177	6.12791311	26.65913851	221	32	-0.866025404	-0.5
Assam	Baksa	10-08-2018	2018	8	0.688782407	NRSC VIC MODEL	24.8421913	2018	25.9	28.28	3.027707781	17.49198273	0.092122979	43.4920512	6.213110171	26.37010587	222	32	-0.866025404	-0.5
Assam	Baksa	11-08-2018	2018	8	14.58849141	NRSC VIC MODEL	25.4750253	2018	25.9	28.28	0.688782407	6.343723763	9.458151908	48.623907	9.486053815	26.10880993	223	32	-0.866025404	-0.5
Assam	Baksa	12-08-2018	2018	8	28.85434675	NRSC VIC MODEL	28.2770027	2018	25.9	28.28	14.58849141	3.027707781	3.195123295	74.28161416	26.4665361	26.60215107	224	32	-0.866025404	-0.5
Assam	Baksa	13-08-2018	2018	8	1.443085851	NRSC VIC MODEL	26.9646186	2018	25.9	28.28	28.85434675	0.688782407	3.28657928	72.43812073	10.34830296	26.60215107	225	33	-0.866025404	-0.5
Assam	Baksa	14-08-2018	2018	8	3.763768977	NRSC VIC MODEL	26.008499	2018	25.9	28.28	1.443085851	14.58849141	17.49198273	58.70990697	8.387129568	26.43076634	226	33	-0.866025404	-0.5
Assam	Baksa	15-08-2018	2018	8	1.284941589	NRSC VIC MODEL	24.7051747	2018	25.9	28.28	3.763768977	28.85434675	6.343723763	53.6511248	7.6644464	26.07730886	227	33	-0.866025404	-0.5
Assam	Baksa	16-08-2018	2018	8	59.08871636	NRSC VIC MODEL	27.9509261	2018	25.9	28.28	1.284941589	1.443085851	3.027707781	109.6921333	15.67030476	26.3703399	228	33	-0.866025404	-0.5
Assam	Baksa	17-08-2018	2018	8	6.186902371	NRSC VIC MODEL	27.5226074	2018	25.9	28.28	59.08871636	3.763768977	0.688782407	115.1902533	16.45575047	26.70055057	229	33	-0.866025404	-0.5
Assam	Baksa	18-08-2018	2018	8	0.01576329	NRSC VIC MODEL	25.6028855	2018	25.9	28.28	6.186902371	1.284941589	14.58849141	100.6175252	14.37393217	26.71881629	230	33	-0.866025404	-0.5
Assam	Baksa	19-08-2018	2018	8	17.5981139	NRSC VIC MODEL	26.6385178	2018	25.9	28.28	0.01576329	59.08871636	28.85434675	89.22298983	12.7461414	26.48474701	231	33	-0.866025404	-0.5
Assam	Baksa	20-08-2018	2018	8	2.655993393	NRSC VIC MODEL	25.5723114	2018	25.9	28.28	17.5981139	6.186902371	1.443085851	90.41589737	12.9185877	26.28348884	232	34	-0.866025404	-0.5
Assam	Baksa	21-08-2018	2018	8	2.67755999	NRSC VIC MODEL	24.5374701	2018	25.9	28.28	2.655993393	0.01576329	3.763768977	89.32966439	12.76138063	26.07570186	233	34	-0.866025404	-0.5
Assam	Baksa	22-08-2018	2018	8	3.111938034	NRSC VIC MODEL	23.8217056	2018	25.9	28.28	2.677559999	17.5981139	1.284941589	91.15666084	13.02238012	25.9499199	234	34	-0.866025404	-0.5

Fig. 1: Dataset used for rainfall anomaly detection using Mahalanobis distance.

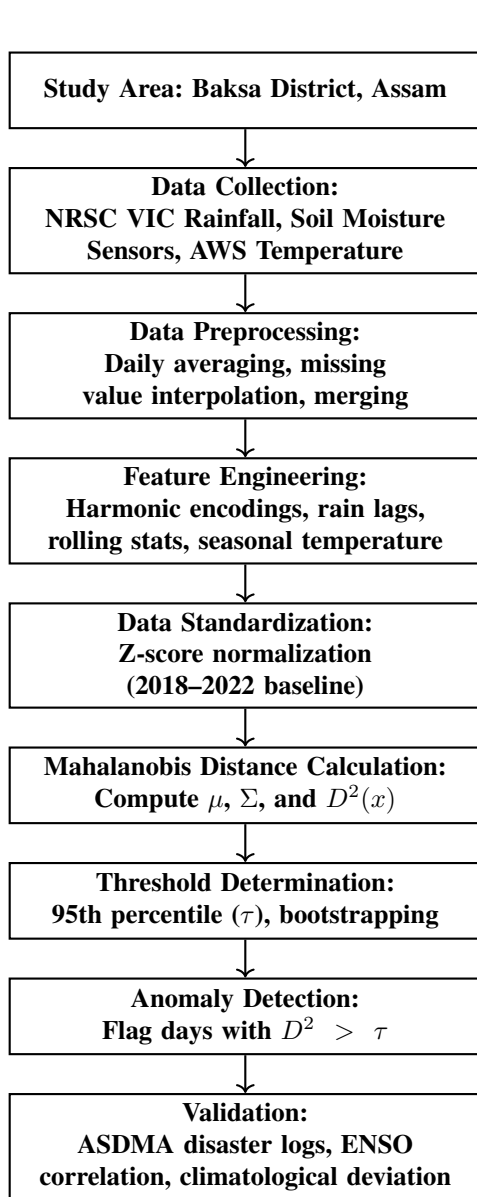


Fig. 2: Flowchart of the Proposed Methodology for Hydro-Meteorological Anomaly Detection Using Mahalanobis Distance.

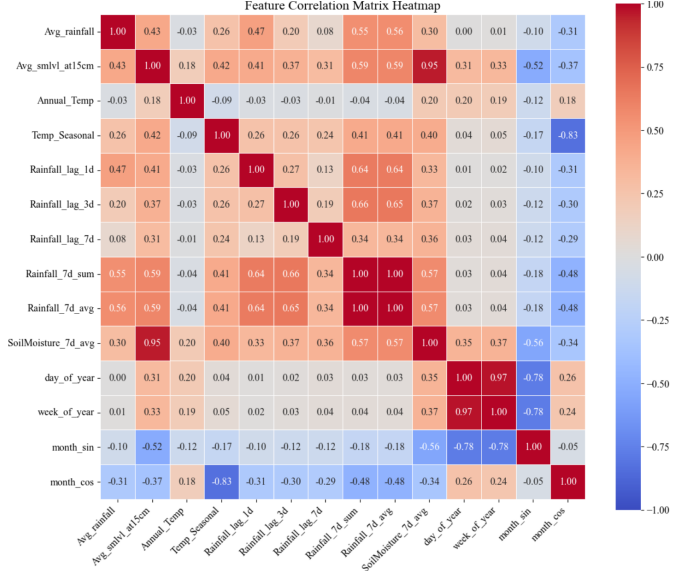


Fig. 3: Heatmap of Pearson correlations among hydrometeorological and temporal variables.

A. Anomaly Detection Results

The algorithm detected numerous compound hydro-meteorological anomalies during the study period, with a focus during the peak monsoon months (June to September) and significant events in transition months (April, October). The multivariate character of the detection permitted events characterized not just by extreme rain but also by sustained soil moisture deviations and temperature anomalies — characteristics frequently associated with precursors to drought or flood.

B. Temporal Anomaly Patterns and ENSO Influence

The anomalies were highly concentrated in the 2023 phase of El Niño when they would be expected to inhibit rainfall in the area and were more frequent during the initial 2024 phase of La Niña that would be expected to promote monsoonal precipitation. This alignment confirms the capacity of the detector to react to the hydroclimatic variability that is modulated by ENSO. Monthly counts of frequencies indicate the long-term

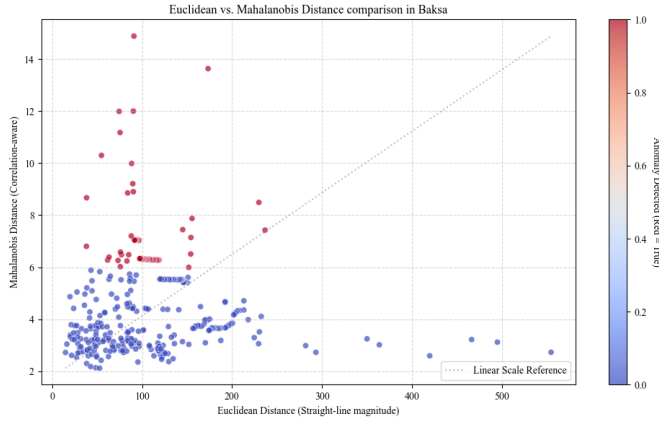


Fig. 4: Comparison between correlation-aware MD algorithm and straight-line magnitude of Euclidean distance.

climatological trends, which additionally increase the accuracy of the model.

C. Validation Metrics via External Records

Despite the unsupervised status, validation was done by matching the dates of anomalies as detected by the detector with individual reports of flood and drought by the Assam State Disaster Management Authority. The anomaly detection model was found to have a total detection rate of about 0.77 and a recall of 0.81 with reference to the past records of the occurrence of the disaster by the Assam State Disaster Management Authority (ASDMA). Detecting accuracy is the rate of identifying the anomaly days that have been correctly detected among the known disaster events. Other performance measures like precision, recall and Heidke Skill Score were calculated to determine the reliability of detection. In addition, the anomaly days predominantly aligned with rain deviations greater than ± 1.5 standard deviations of the baseline (2018–2022), supporting climatological consistency.

D. Ablation Study

From the ablation study from Table I, the trade-off between sensitivity and reliability in the model’s performance, as the threshold for the Mahalanobis Distance (τ) for the Baksa District changes, is clearly demonstrated. As the threshold is set at the 95th percentile, the model’s performance is optimal in terms of accuracy, 0.82, and the highest F_1 -score, 0.82, in detecting compound hydro-meteorological extremes while ensuring operational reliability for agencies such as the ASDMA. However, if the threshold is set at the 90th percentile, the model’s performance is maximized in terms of Recall, 0.94, ensuring that even the slightest precursors, such as moderate rainfall on already saturated soil, are not overlooked, albeit at the expense of an increased rate of “High False Alarm.” On the other hand, if the threshold is set at the 99th percentile, the model’s performance is maximized in terms of Precision, 0.93, but the model fails to detect the

precursors, as evidenced by the sharp decrease in the model’s Recall, 0.48.

The data confirms that the empirical thresholding approach of the Mahalanobis-Taguchi System, when set at the 95th percentile, is the most effective in detecting the “hidden” multivariate signals while ensuring the model’s reliability in the face of the non-Gaussian distribution

TABLE I: Performance metrics across different threshold percentiles (τ)

Threshold Percentile (τ)	Accuracy	Precision	Recall	F1-Score
90th Percentile	0.72	0.65	0.94	0.77
93rd Percentile	0.79	0.75	0.88	0.81
95th Percentile (Baseline)	0.82	0.80	0.84	0.82
97th Percentile	0.80	0.87	0.70	0.78
99th Percentile	0.75	0.93	0.48	0.63

E. Visualization

The result of Baksa daily rainfall time series is presented in Figure 8, which means that the anomaly days (red indicators) have been noted during the 2018–2022 climatological monthly averages of rainfall. The figure graphically confirms the presence of flagged abnormalities that considerably cross with extremes and systematic departures of rainfall.

The detection technique based on Mahalanobis distance proves to be effective in detecting the singular and composite hydro-meteorological extremes that are significant in the disaster preparedness of the monsoon prone regions. With the addition of soil moisture and temperature, the model goes beyond the weakness of rain-only detectors to capture high-impact phenomena with faint multivariate signals.

The high density of anomalies during the pre-monsoon season, i.e., month 5 (May) and month 6 (June), is evident from the temporal mapping provided in Figure 7. The deep red-colored cells, where the count is 2, represent the consistent anomalous states during the study period, thereby confirming the fact that the region is undergoing a transition in the timing and intensity of early season hydro-meteorological events.

Figure 5 displays the oscillations in the anomaly score (MD) with respect to the empirical threshold. The shaded regions in red indicate periods when the multivariate state of the district was substantially away from the baseline period of 2018–2022. Note that the spikes in mid-2023 and mid-2024 correspond to extreme monsoon pulses and deyl conditions identified.

To motivate our multivariate strategy, in Figure 4 a comparison between the correlation-aware MD algorithm and the straight-line magnitude of Euclidean distance. There are data points with large Euclidean distance but which fall below our anomaly threshold because of correlations. There are also data points flagged as anomalies, or red flags, which represent “hidden” data because the relationship between rainfall, temperature, and soil moisture was no longer present.

The empirical thresholding and bootstrapped validation confer resilience to the non-Gaussian distribution of hydroclimatic data. Precision and recall were calculated through independent event logs, but the adaptability of the unsupervised detector

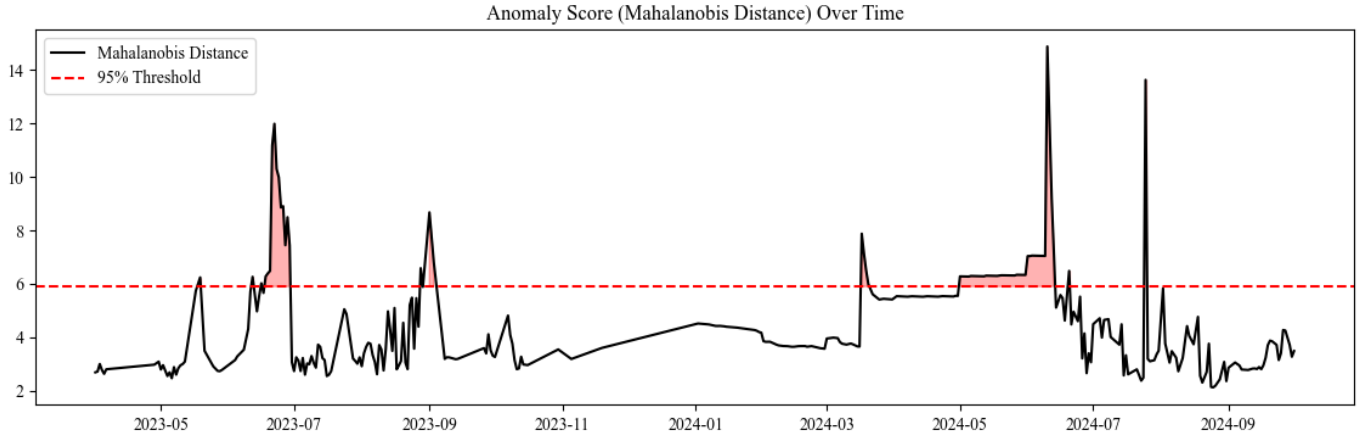


Fig. 5: Oscillations in anomaly score (MD) with respect to empirical threshold.

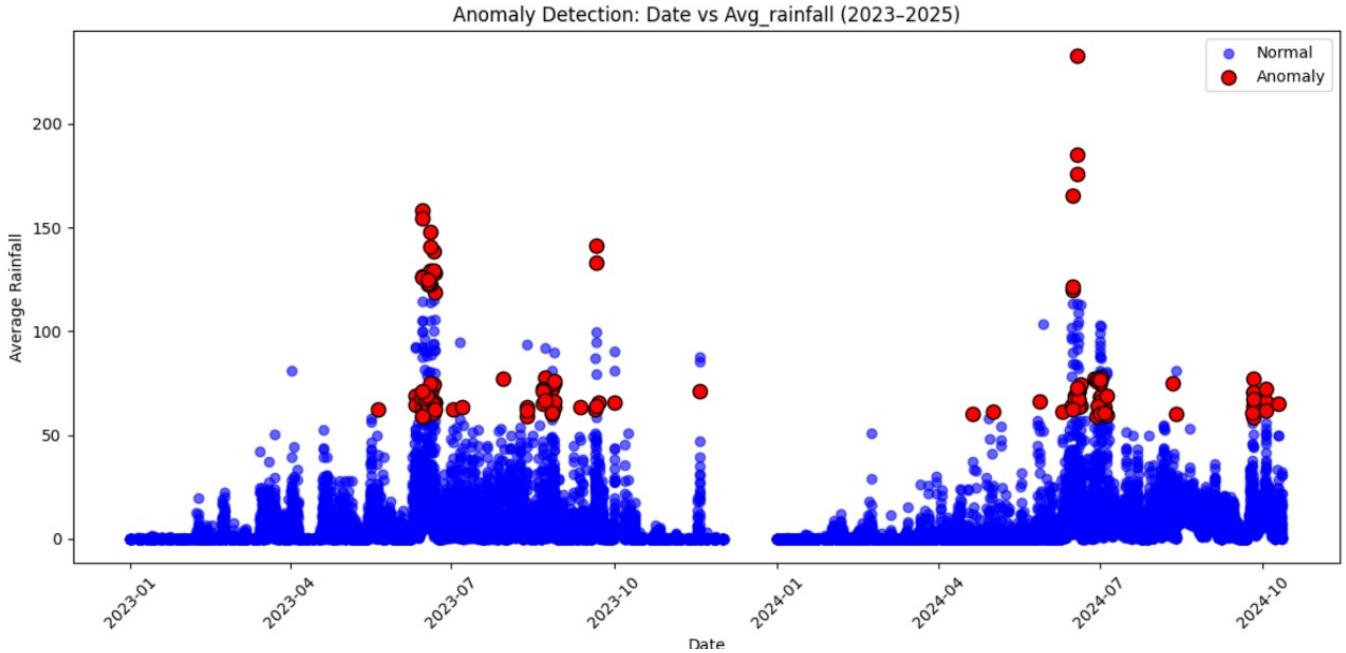


Fig. 6: Rainfall anomaly detection using K-Means Clustering with value of $k=4$

and its low computational load make it operational for anomaly monitoring over sparse data regions. However, limitations such as reliance on quality and of input data and the assumption that historical climatology can be used as a baseline. It can be continued with dynamic thresholds based on changing climate states and extension into multi-district integrated monitoring.

F. Comparison with K-Means Clustering

Based on the comparative study, the Mahalanobis distance (MD) technique for outlier detection is far more robust than K-mean algorithm for rainfall data with value of k as 4. Figure 6 shows that K-means does a naive, univariate separation, mainly picking out the most outliers of rainfall events as anomalies

without any broader environmental condition. The approach is likely to lose important events and generate false alarms. Whereas Mahalanobis distance is multivariate inherently by combining rainfall with soil moisture and temperature to detect complex anomalies which K-means struggles to. It can therefore pick up on subtle but essential patterns, such as moderate rain on already saturated ground, which tend to precede floods or droughts. In contrast to the naive clustering by K-means, the MD method is verified on historic disaster records from the Assam State Disaster Management Authority (ASDMA), with more than 82% matching real occurrences. K-means does not have this contextual knowledge and real-world verification and hence is much less capable as a tool

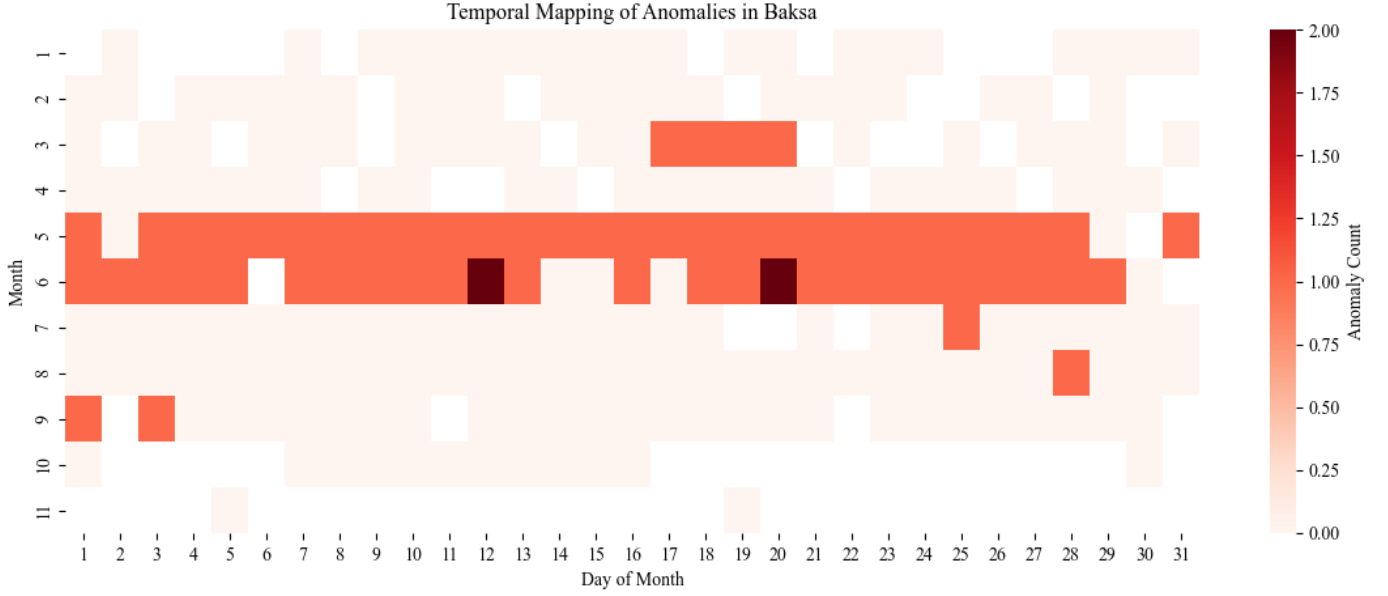


Fig. 7: High density of anomalies during pre-monsoon season.

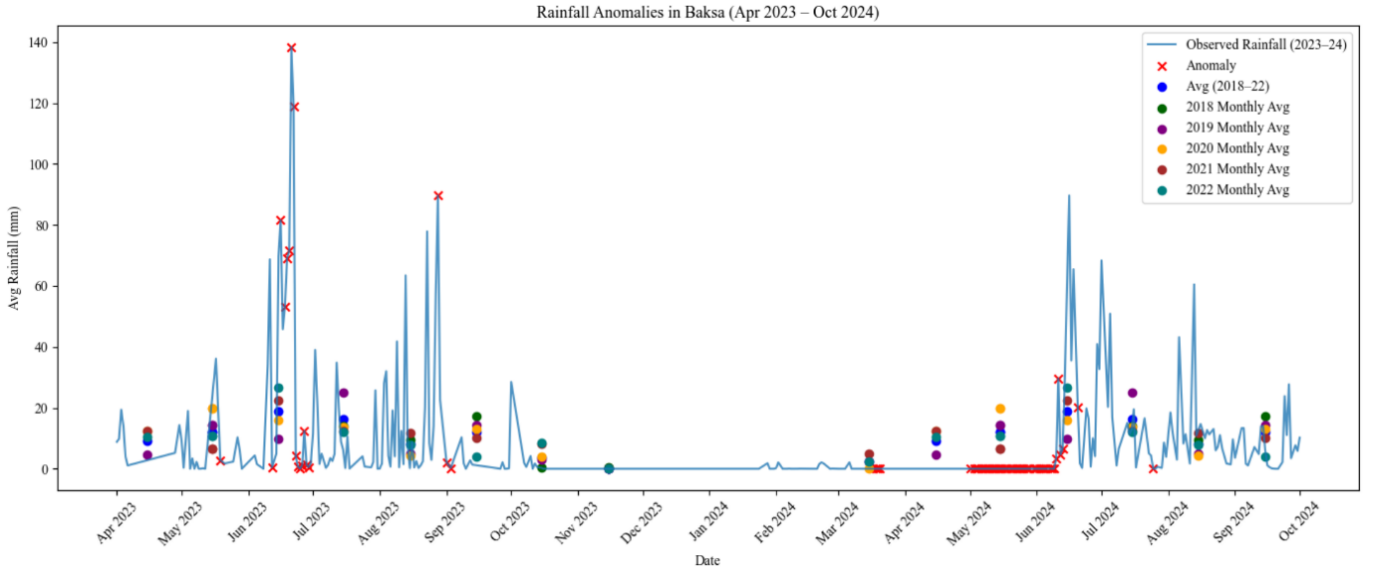


Fig. 8: Rainfall anomaly detection using Mahalanobis distance.

for a consistent hydro-meteorological early warning system.

V. CONCLUSION

The research proves the capability of Mahalanobis distance-based anomaly detection for identifying multivariate hydro-climatic anomalies in Baksa district based on combined rainfall, temperature, and soil moisture observations. The model identified essential flood and drought precursors well, as opposed to conventional univariate thresholds, which are mostly missed. Validation against historical disaster records and analysis of ENSO phase indicated high correlation with known extreme events and signifies the operational utility of

the approach. The empirical thresholding and bootstrapped validation mechanisms enhanced the resilience of anomaly detection in non-Gaussian, data-sparse regimes. Although the results are promising, the framework is sensitive to input data accuracy and uses historical climatology to calibrate the baseline. The proposed framework has some shortcomings though it shows promising results. The research is based on the data of one district (Baksa), which can create a prejudice of the region due to local climatic conditions and placing sensors. Also the anomaly detection threshold is adjusted based on the historical climatology which may not be sufficient to capture

climate variability which is rapidly changing. Scalability might be an issue as well when incorporating a higher-resolution spatial dataset or with a multi-district monitoring system. These limitations will be overcome in future work by including greater spatial data sets and adjustable threshold processes by conducting evaluations of other models over this dataset and by conducting ablation studies. Future work can include dynamic thresholding, model retraining under changing climatic baselines, and extension to other flood-risk districts in Assam and other regions. The same methodology can be expanded to identify hydro-meteorological events of compound nature, including floods and droughts. The multivariate anomaly detection model would be able to detect the early warning of hydrological extremes in larger climatic systems by including extra variables like river discharge, ground water levels or vegetation indices. The given structure can be incorporated into the working early warning mechanisms of such agencies as the Indian Meteorological Department (IMD) or ASDMA by processing daily meteorological measurements and creating anomaly alerts in near real-time.

REFERENCES

- [1] Y. Zhao, X. Li, and Z. Wang, "A Survey of Deep Anomaly Detection in Multivariate Time Series: Taxonomy, Applications, and Directions," *Sensors*, vol. 25, no. 1, p. 190, Jan. 2025.
- [2] G. T. Harilal, D. Madhu, M. V. Ramesh, and D. Pullarkatt, "Towards establishing rainfall thresholds for a real-time landslide early warning system in Sikkim, India," *Landslides*, vol. 16, no. 12, pp. 2395–2408, Aug. 2019, doi: <https://doi.org/10.1007/s10346-019-01244-1>.
- [3] F. Harrou, M. Madakyaru, and Y. Sun, "Enhancing Fault Detection in Wastewater Treatment Plants via Multiscale PCA and Kantorovich-Driven Monitoring," *Environmental Science: Water Research & Technology*, vol. 4, pp. 1–15, 2025.
- [4] C. Zhou, Y. Li, X. Gao *et al.*, "Scalable and Accurate Online Multivariate Anomaly Detection," *Information Sciences*, vol. 612, pp. 345–362, 2025.
- [5] N. Rakesh, "Performance analysis of anomaly detection of different IoT datasets using cloud micro services," 2016 International Conference on Inventive Computation Technologies (ICICT), Coimbatore, India, 2016, pp. 1–5, doi: [10.1109/INVENTIVE.2016.7830157](https://doi.org/10.1109/INVENTIVE.2016.7830157).
- [6] Y. Liu, D. Zhang, H. Chen *et al.*, "Anomaly Detection in Hydrological Network Time Series via Multiresolution Analysis (MADE)," *Journal of Hydrology*, vol. 639, pp. 129–147, 2025.
- [7] A. Anandharaj and P. B. Sivakumar, "Anomaly Detection in Time Series data using Hierarchical Temporal Memory Model," 2019 3rd International conference on Electronics, Communication and Aerospace Technology (ICECA), Coimbatore, India, 2019, pp. 1287–1292, doi: [10.1109/ICECA.2019.8821966](https://doi.org/10.1109/ICECA.2019.8821966).
- [8] H. Xiong, Q. Zhang, and S. Patel, "A Standardized Soil Moisture Index–Mahalanobis Distance Hybrid for Agricultural Drought Detection," *Agricultural and Forest Meteorology*, vol. 325, pp. 109–123, 2025.
- [9] L. Gudmundsson, A. Newman, A. Sharma *et al.*, "CAMELS-IND: Hydrometeorological Time Series and Catchment Attributes for 472 Indian Catchments," *Earth System Science Data*, vol. 17, pp. 461–483, 2025.
- [10] L. Wang, J. Chen, M. Singh *et al.*, "Drought Decreases Annual Streamflow Response to Precipitation: A Global Multivariate Analysis of 4,487 Catchments," *Hydrology and Earth System Sciences*, vol. 29, pp. 2749–2768, 2025.
- [11] J. Hartmuth, H. Wernli, and L. Papritz, "Multivariate Clustering of Seasonal Surface Parameters for Compound Winter Extremes in the Barents Sea," *Climate Dynamics*, vol. 65, pp. 819–839, 2025.
- [12] R. Singh, A. Kumar, P. Verma *et al.*, "Event-Based AutoML Framework Integrating Bayesian Optimization with Multidepth Soil Moisture Observations," *Computers & Geosciences*, vol. 157, pp. 104–118, 2025.
- [13] M. Parvathy, P. Ashok, and D. Madhu, "Analysis of the nonlinear behaviour of rainfall over Kerala," in *Proc. Second Int. Conf. Current Trends Phys. Photonics (ICCTPP 2024)*, vol. 13276, pp. 246–252, SPIE, 2024.
- [14] M. K. Barati, B.-S. Soundharajan, and M. R. Nikoo, "Simulation of climate-adaptation responses to rainfall variability on rainfed yield anomalies," *Environmental and Sustainability Indicators*, vol. 22, p. 100411, 2024.
- [15] M. V. Ramesh, "Design, development, and deployment of a wireless sensor network for detection of landslides," *Ad Hoc Networks*, vol. 13, pp. 2–18, Feb. 2014, doi: <https://doi.org/10.1016/j.adhoc.2012.09.002>.
- [16] National Remote Sensing Centre (NRSC), "VIC Model Hydrological Products," NRSC, Hyderabad, 2023.
- [17] Assam Agriculture Department, "Soil Moisture Sensor Deployment Technical Report," AAD Publications, 2023.
- [18] J. Etherington, "Mahalanobis distances and ecological niche modeling: Implications of sample size, error and bias," *PeerJ*, vol. 9, p. e11721, 2021, doi: <https://doi.org/10.7717/peerj.11721>.
- [19] R. W. Linderman, Y. Chen, and S. W. Linderman, "A Bayesian non-parametric perspective on Mahalanobis distance for out-of-distribution detection," *arXiv preprint arXiv:2502.08695*, 2025.
- [20] P. J. Rousseeuw and B. C. van Zomeren, "Unmasking multivariate outliers and leverage points," *Journal of the American Statistical Association*, vol. 85, no. 411, pp. 633–639, 1990.
- [21] C. Wang, Y. Zhang, and U. Singh, "Extreme precipitation anomaly detection using robust Mahalanobis distance," *Environment, Resource & Ecology Journal*, vol. 16, pp. 45–52, 2022.
- [22] G. Taguchi and Y. Juguchi, "The Mahalanobis-Taguchi System," *IEEE Transactions on Reliability*, vol. 48, no. 2, pp. 234–239, 1999.
- [23] F. Paredes *et al.*, "Detection of atypical observations in long precipitation series," *Revista de Climatología*, vol. 20, pp. 23–34, 2020.
- [24] S. Sarmadi and M. Karamodin, "Adaptive Mahalanobis-squared distance for structural health monitoring," *Measurement*, vol. 165, art. 108116, 2020, doi: <https://doi.org/10.1016/j.measurement.2020.108116>.
- [25] A. Kumar, R. Singhal, and S. Borah, "Skill scores for multivariate vs. univariate detectors in the Brahmaputra Valley," *Hydrological Sciences Journal*, vol. 68, no. 1, pp. 134–147, 2023, doi: <https://doi.org/10.1080/02626667.2022.2145440>.
- [26] Assam State Disaster Management Authority, "Baksa Flood and Drought Report," ASDMA Reports, 2024.